MAYUR GANGIREDDY

AI/ML Engineer • Full-Stack Systems Developer

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PROFESSIONAL SUMMARY

AI/ML & Full-Stack Engineer with proven production experience deploying 5-second interval time-series pipelines, edge intelligence, and cross-platform mobile systems published to the Google Play Store and Apple App Store. Specialized in energy disaggregation (NILM), privacy-preserving Model Context Protocol (MCP) data proxies, native hardware integrations (USB IRDA / OCR), and high-throughput web microservices.

WORK EXPERIENCE

Bharat Smart Services – **Machine Learning & Full-Stack Engineer** 01/2026 – Present | Hyderabad, India

- **NILM Time-Series Energy Disaggregation:** Built production appliance disaggregation models on 5-second interval smart meter telemetry. Benchmarked 1D/2D CNNs and Temporal Transformers against an optimized sliding-window **XGBoost** pipeline; achieved **96% F1-score on Geysers** and **~85% F1-score across AC, Refrigerator, and Washing Machines** (evaluated on held-out temporal splits), yielding sub-10ms CPU inference latency.
- **NerveNet – Client-Side Encrypted Enterprise MCP Platform:** Architected an enterprise AI agent platform on the **Model Context Protocol (MCP)**. Engineered client-side AES-256 encryption and key tokenization over private CSV tables and SQL databases, allowing cloud LLMs to execute reasoning and queries strictly over anonymized tokens without exposing unencrypted corporate records.
- **Bharat Smart Services App (Utility & Billing)** Google Play App Store: Engineered cross-platform React/Capacitor app for live state discom billing (TSSPDCL, TSNPDCL, HMWSSB). Built custom native Java USB serial drivers for optical IRDA dongles, camera-based OCR meter reading review, and native iOS/Android home screen widgets.
- **Bharat Smart Home (IoT Automation & Telemetry)** Google Play App Store: Architected smart home platform featuring hierarchical device control, condition-based automation schedules, lifestyle scene orchestration, and real-time disaggregated appliance energy intelligence.

Telangana State Police Academy (TGPA) – **Android Application Developer** 03/2024 – 08/2024 | Hyderabad, India

- **Criminal Laws by TGPA** Google Play Store: Sole developer of official state police legal repository app published on Google Play. Built sub-50ms offline search over criminal penal codes using SQLite FTS, bilingual legal document viewers, and AES-encrypted local storage.

Acmegrade – **Android App Development Intern** 11/2023 – 01/2024 | Remote

- Developed and deployed native Android mobile application modules in Java, implementing activity lifecycles, UI layout hierarchies, REST API integration, and local SQLite caching.

SELECTED PROJECTS

- **Hostel Hub Pro & Companion Mobile App:** Multi-tenant accommodation platform connecting administrators, wardens, students, and guards. Built administrative web portal with **TanStack Start SSR & React**, companion **Flutter app** featuring time-restricted (24-hr) HMAC dynamic QR gate passes with server-side nonce verification to prevent pass re-use and sharing, and an Express/MySQL backend on Plesk.
- **Satellite Ship Detection & Classification (YOLOv8)** Live Demo GitHub: Engineered computer vision pipeline for high-resolution optical satellite imagery analysis. Implemented automated water/land masking followed by fine-tuned YOLOv8 deep learning vessel detection with confidence-scored bounding boxes.
- **Samarth – Intelligent Agriculture & Climate Q&A Engine** GitHub: Built Natural Language to SQL analytics engine over Indian agricultural datasets using FastAPI and DuckDB. Enforced schema-constrained LLM prompt architecture and AST query validation, guaranteeing 100% auditable generated SQL execution logs.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript (ES6+), Java, SQL, Dart, C, C++

AI & Machine Learning: XGBoost, PyTorch, Transformers, CNNs, Time-Series NILM, Model Context Protocol (MCP), YOLOv8, Computer Vision (OCR), LangChain, Scikit-Learn

Web & Mobile: React.js, Node.js, Express, FastAPI, Capacitor.js, Flutter, Android (Java/Kotlin), Tailwind CSS, Vite

Databases & Cloud: PostgreSQL, MySQL, MariaDB, DuckDB, Supabase, Firebase, Plesk Linux, RESTful APIs

Tools & DevOps: Git, Docker, Android Studio, Xcode, Postman, Linux CLI, USB Serial Hardware (IRDA)

EDUCATION & HONORS

Sphoorthy Engineering College – **B.Tech in Computer Science & Engineering (AIML)** 2021 – 2025 | Hyderabad, India

Sri Gayatri Junior College – **Board of Intermediate Education (MPC)** 2019 – 2021 | Hyderabad, India

National Finalist – Smart India Hackathon (SIH 2024)

Production Apps – 3 Published Apps on Google Play Store & Apple App Store